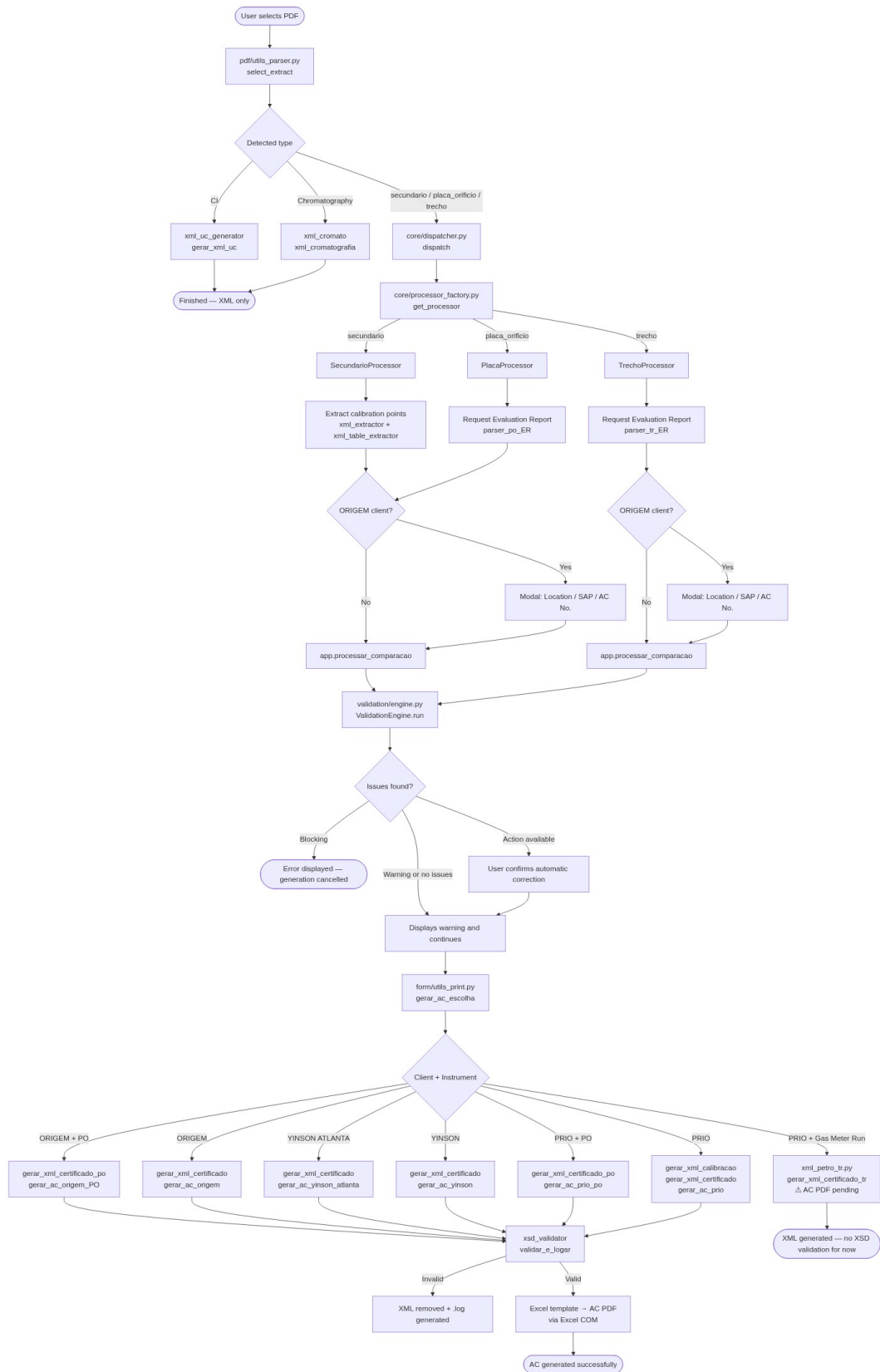


AC's Generator

Technical Documentation — Main Flow · Rules · Clients

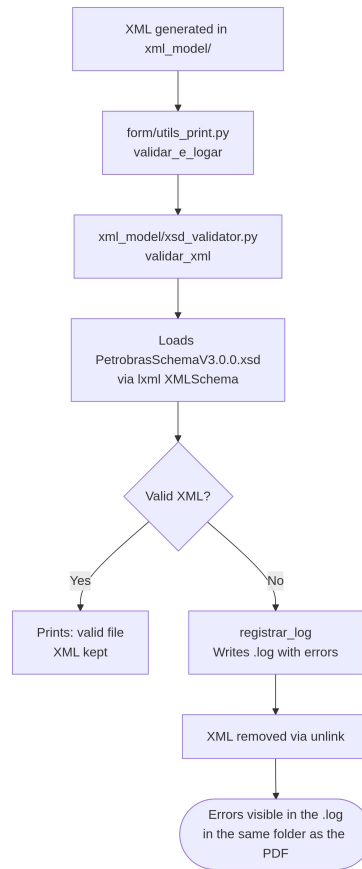
1. Main Flow — PDF → AC + XML

Starting from a PDF calibration certificate, the system detects the instrument type, extracts the data, validates it against the database and the XSD schemas, and generates the Petrobras XML along with the Critical Analysis (AC) PDF.



2. XSD Validation Flow

After the XML is generated, the file is validated against the *PetrobrasSchemaV3.0.0.xsd* schema. If an error occurs, the XML is removed and a .log file is generated in the same folder as the PDF.



Note: the Gas Meter Run flow does not yet go through XSD validation — the XML is generated directly for testing.

3. Validation Rules — Secondary Instruments

The validation engine (validation/engine.py) runs the rules below. Each rule can return a warning, a corrective action, or a generation block.

Rule	What it checks
regra_tag_vs_sn	Mismatch between TAG and SN (MVS detection)
regra_novo_instrumento	Instrument not registered → offers automatic insertion
regra_sn_instrumento	Instrument SN differs from the registered record
regra_sn_sensor	Sensor SN differs from the registered record
regra_range	Calibration range differs from the registered record
regra_haste_te	TE sensor rod validation
regra_local_fpso	Inconsistent FPSO location
regra_rangein	Indicated range vs. calibration range
regra_incert_fidu	Uncertainty / fiducial error out of limit
regra_cmc	CMC (Capability Measurement Capability)
regra_classe	Instrument class
data_proxcal	Next calibration date
prazo_emissao	Certificate issuance deadline

4. Supported Clients and Templates

Each client × instrument combination has a dedicated Excel template and a specific XML generator. The *AC PDF* column indicates whether the PDF generation is implemented (✓) or pending.

Client	Instrument	Excel Template	XML Generator	AC PDF
ORIGEM Energia Alagoas	Secondary	TemplateAC_ORIGEM.xlsx	xml_petro_generator.py	✓
ORIGEM Energia Alagoas	Orifice Plate	TemplateAC_PO_ORIGEM.xlsx	xml_petro_po.py	✓
PRIO	Secondary	TemplateAC_PRIO.xlsx	xml_generator.py + xml_petro_generator.py	✓
PRIO	Orifice Plate	TemplateAC_PO_PRIO.xlsx	xml_petro_po.py	✓
PRIO	Gas Meter Run	—	xml_petro_tr.py	<i>pending</i>
YINSON	Secondary	TemplateAC_YINSON.xlsx	xml_petro_generator.py	✓
YINSON (FPSO Atlanta)	Secondary	TemplateAC_YINSON - ATLANTA.xlsx	xml_petro_generator.py	✓